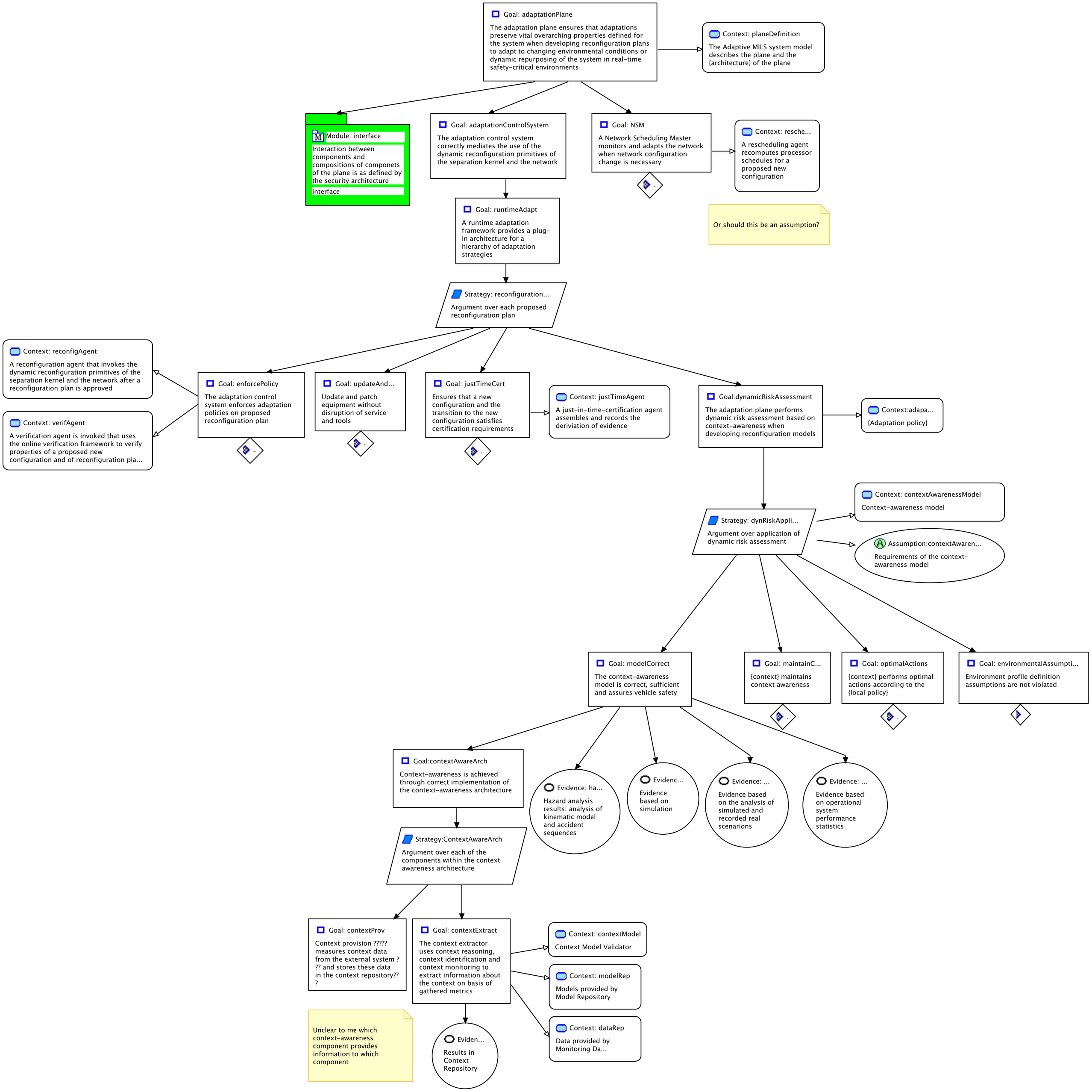
Goal: adaptationPlane
The adaptation plane ensures that adaptations preserve vital overarching properties defined for the system when developing reconfiguration plans to adapt to changing environmental conditions or dynamic repurposing of the system in real-time safety-critical environments
Context: planeDefinition
The Adaptive MILS system model describes the plane and the {architecture} of the plane
Module: interface
Interaction between components and compositions of componets of the plane is as defined by the security architecture
interface
Goal: adaptationControlSystem
The adaptation control system correctly mediates the use of the dynamic reconfiguration primitives of the separation kernel and the network
Goal: NSM
A Network Scheduling Master monitors and adapts the network when network configuration change is necessary
Context: resche...
A rescheduling agent recomputes processor schedules for a proposed new configuration
Or should this be an assumption?
Goal: runtimeAdapt
A runtime adaptation framework provides a plug-in architecture for a hierarchy of adaptation strategies
Strategy: reconfiguration...
Argument over each proposed reconfiguration plan
Context: reconfigAgent
A reconfiguration agent that invokes the dynamic reconfiguration primitives of the separation kernel and the network after a reconfiguration plan is approved
Context: verifAgent
A verification agent is invoked that uses the online verification framework to verify properties of a proposed new configuration and of reconfiguration pla...
Goal: enforcePolicy
The adaptation control system enforces adaptation policies on proposed reconfiguration plan
Goal: updateAnd...
Update and patch equipment without disruption of service and tools
Goal: justTimeCert
Ensures that a new configuration and the transition to the new configuration satisfies certification requirements
Context: justTimeAgent
A just-in-time-certification agent assembles and records the deriviation of evidence
Goal:dynamicRiskAssessment
The adaptation plane performs dynamic risk assessment based on context-awareness when developing reconfiguration models
Context:adapa...
{Adaptation policy}
Context: contextAwarenessModel
Context-awareness model
Strategy: dynRiskAppli...
Argument over application of dynamic risk assessment
Assumption:contextAwaren...
Requirements of the context-awareness model
Goal: modelCorrect
The context-awareness model is correct, sufficient and assures vehicle safety
Goal: maintainC...
{context} maintains context awareness
Goal: optimalActions
{context} performs optimal actions according to the {local policy}
Goal: environmentalAssumpti...
Environment profile definition assumptions are not violated
Goal:contextAwareArch
Context-awareness is achieved through correct implementation of the context-awareness architecture
Evidence: ha...
Hazard analysis results: analysis of kinematic model and accident sequences
Evidenc...
Evidence based on simulation
Evidence: ...
Evidence based on the analysis of simulated and recorded real scenarios
Evidence: ...
Evidence based on operational system performance statistics
Strategy:ContextAwareArch
Argument over each of the components within the context awareness architecture
Goal: contextProv
Context provision ????? measures context data from the external system ?? ?? and stores these data in the context repository?? ?
Goal: contextExtract
The context extractor uses context reasoning, context identification and context monitoring to extract information about the context on basis of gathered metrics
Context: contextModel
Context Model Validator
Context: modelRep
Models provided by Model Repository
Context: dataRep
Data provided by Monitoring Da...
Unclear to me which context-awareness component provides information to which component
Eviden...
Results in Context Repository